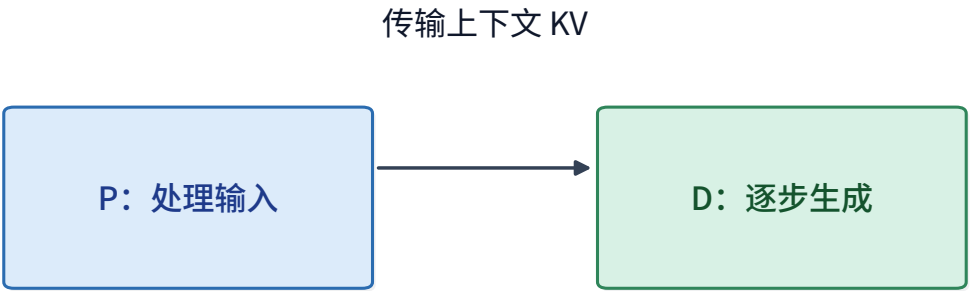PD：按输入处理与生成阶段分工